LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various causes of aircraft wiring degradation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various causes of aircraft wiring degradation.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams showing wiring degradation types; Instructor supervision required for safety analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft wiring?
(b) List three common materials used in aircraft wire insulation.
(c) Why is wiring degradation a critical safety concern?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Environmental factors causing degradation • Mechanical stress and vibration effects • Chemical corrosion mechanisms • Thermal cycling impacts on insulation • Improper installation practices	L & Ds	PPT, IB	15 min
(b)	• Analysis of chafing and abrasion points • Effects of moisture and humidity ingress • Impact of oil and fuel contamination • UV radiation damage to outer sheaths • Consequences of improper termination	L & Ds	PPT, PDF	20 min
(c)	• Preventive maintenance strategies • Inspection intervals for wiring systems • Replacement criteria for degraded wires • Documentation of wiring defects • Role of lubricants in protection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how vibration contributes to wiring degradation.
(b) Describe the relationship between moisture ingress and electrical failure.
(c) Identify two signs of chemical corrosion in aircraft wiring.

12. Home Assignment and Follow Up	Prepare a brief note on how the topics covered in this lesson (wiring degradation causes) relate to the load protection techniques discussed in LP 03.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Electrical load determination and various load protection techniques.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min